Step 6. Write the partial products, lining up the digits in the place values with the numbers above. Repeat for the tens place in the bottom number, the hundreds place, and so on.
Step 7. Insert a zero as a placeholder with each additional partial product.
Step 8. Add the partial products.

1.5 Divide Whole Numbers

Operation	Notation	Expression	Read as	Result
Division	$\div$ $\frac{a}{b}$ $b\overline{)a}$ a/b	$12 \div 4$ $\frac{12}{4}$ $4\overline{)12}$ 12/4	Twelve divided by four	the quotient of 12 and 4

- **Division Properties of One**
  - Any number (except 0) divided by itself is one. $a \div a = 1$
  - Any number divided by one is the same number. $a \div 1 = a$
- **Division Properties of Zero**
  - Zero divided by any number is 0. $0 \div a = 0$
  - Dividing a number by zero is undefined. $a \div 0$ undefined
- **Divide whole numbers.**
  Step 1. Divide the first digit of the dividend by the divisor.
  If the divisor is larger than the first digit of the dividend, divide the first two digits of the dividend by the divisor, and so on.
  Step 2. Write the quotient above the dividend.
  Step 3. Multiply the quotient by the divisor and write the product under the dividend.
  Step 4. Subtract that product from the dividend.
  Step 5. Bring down the next digit of the dividend.
  Step 6. Repeat from Step 1 until there are no more digits in the dividend to bring down.
  Step 7. Check by multiplying the quotient times the divisor.

Exercises

Review Exercises

Introduction to Whole Numbers

Identify Counting Numbers and Whole Numbers

In the following exercises, determine which of the following are (a) counting numbers (b) whole numbers.

453. $0, 2, 99$

454. $0, 3, 25$

455. $0, 4, 90$

456. $0, 1, 75$

Model Whole Numbers

In the following exercises, model each number using base-10 *blocks and then show its value using place value notation.*

457. 258

458. 104

Identify the Place Value of a Digit

In the following exercises, find the place value of the given digits.

459. 472,981
ⓐ 8 ⓑ 4 ⓒ 1 ⓓ 7
ⓔ 2

460. 12,403,295
ⓐ 4 ⓑ 0 ⓒ 1 ⓓ 9
ⓔ 3

Use Place Value to Name Whole Numbers

In the following exercises, name each number in words.

461. 5,280

462. 204,614

463. 5,012,582

464. 31,640,976

Use Place Value to Write Whole Numbers

In the following exercises, write as a whole number using digits.

465. six hundred two

466. fifteen thousand, two hundred fifty-three

467. three hundred forty million, nine hundred twelve thousand, sixty-one

468. two billion, four hundred ninety-two million, seven hundred eleven thousand, two

Round Whole Numbers

In the following exercises, round to the nearest ten.

469. 412

470. 648

471. 3,556

472. 2,734

In the following exercises, round to the nearest hundred.

473. 38,975

474. 26,849

475. 81,486

476. 75,992

Add Whole Numbers

Use Addition Notation

In the following exercises, translate the following from math notation to words.

477. $4 + 3$

478. $25 + 18$

479. $571 + 629$

480. $10{,}085 + 3{,}492$

Model Addition of Whole Numbers

In the following exercises, model the addition.

481. $6 + 7$

482. $38 + 14$

Add Whole Numbers

In the following exercises, fill in the missing values in each chart.

483.

+	0	1	2	3	4	5	6	7	8	9
0	0	1		3	4		6	7		9
1	1	2	3	4			7	8	9	
2		3	4	5	6	7	8		10	11
3	3		5		7	8		10		12
4	4	5			8	9			12	
5	5		7	8			11		13	
6	6	7	8		10			13		15
7			9			12	13		15	16
8	8	9		11			14		16	
9	9	10	11		13	14			17	

484.

+	3	4	5	6	7	8	9
6							
7							
8							
9							

In the following exercises, add.

485. ⓐ $0 + 19$ ⓑ $19 + 0$

486. ⓐ $0 + 480$ ⓑ $480 + 0$

487. ⓐ $7 + 6$ ⓑ $6 + 7$

488. ⓐ $23 + 18$ ⓑ $18 + 23$

489. $44 + 35$

490. $63 + 29$

491. $96 + 58$

492. $375 + 591$

493. $7{,}281 + 12{,}546$

494. $5{,}280 + 16{,}324 + 9{,}731$

Translate Word Phrases to Math Notation

In the following exercises, translate each phrase into math notation and then simplify.

495. the sum of 30 and 12

496. 11 increased by 8

497. 25 more than 39

498. total of 15 and 50

Add Whole Numbers in Applications

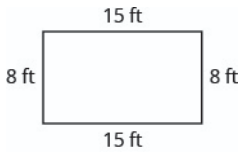
In the following exercises, solve.

499. **Shopping for an interview** Nathan bought a new shirt, tie, and slacks to wear to a job interview. The shirt cost $24, the tie cost $14, and the slacks cost $38. What was Nathan's total cost?

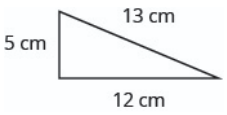
500. **Running** Jackson ran 4 miles on Monday, 12 miles on Tuesday, 1 mile on Wednesday, 8 miles on Thursday, and 5 miles on Friday. What was the total number of miles Jackson ran?

In the following exercises, find the perimeter of each figure.

501.



502.



Subtract Whole Numbers

Use Subtraction Notation

In the following exercises, translate the following from math notation to words.

503. $14 - 5$

504. $40 - 15$

505. $351 - 249$

506. $5{,}724 - 2{,}918$

Model Subtraction of Whole Numbers

In the following exercises, model the subtraction.

507. $18 - 4$

508. $41 - 29$

Subtract Whole Numbers

In the following exercises, subtract and then check by adding.

509. $8 - 5$

510. $12 - 7$

511. $23 - 9$

512. $46 - 21$

513. $82 - 59$

514. $110 - 87$

515. $539 - 217$

516. $415 - 296$

517. $1{,}020 - 640$

518. $8{,}355 - 3{,}947$

519. $10{,}000 - 15$

520. $54{,}925 - 35{,}647$

Translate Word Phrases to Math Notation

In the following exercises, translate and simplify.

521. the difference of nineteen and thirteen

522. subtract sixty-five from one hundred

523. seventy-four decreased by eight

524. twenty-three less than forty-one

Subtract Whole Numbers in Applications

In the following exercises, solve.

525. **Temperature** The high temperature in Peoria one day was 86 degrees Fahrenheit and the low temperature was 28 degrees Fahrenheit. What was the difference between the high and low temperatures?

526. **Savings** Lynn wants to go on a cruise that costs $\$2{,}485$. She has $\$948$ in her vacation savings account. How much more does she need to save in order to pay for the cruise?

Multiply Whole Numbers

Use Multiplication Notation

In the following exercises, translate from math notation to words.

527. 8×5

528. $6 \cdot 14$

529. $(10)(95)$

530. $54(72)$

Model Multiplication of Whole Numbers

In the following exercises, model the multiplication.

531. 2×4

532. 3×8

Multiply Whole Numbers

In the following exercises, fill in the missing values in each chart.

533.

×	0	1	2	3	4	5	6	7	8	9
0	0	0	0	0	0		0		0	0
1	0	1	2		4	5	6	7		9
2	0		4		8	10		14	16	
3		3		9			18		24	
4	0	4		12			24			36
5	0	5	10		20		30	35	40	45
6			12	18			36	42		54
7	0	7		21		35			56	63
8	0	8	16		32		48		64	
9			18	27	36			63	72	

534.

×	3	4	5	6	7	8	9
6							
7							
8							
9							

In the following exercises, multiply.

535. $0 \cdot 14$

536. $(256)0$

537. $1 \cdot 99$

538. $(4{,}789)1$

539. ⓐ $7 \cdot 4$ ⓑ $4 \cdot 7$

540. $(25)(6)$

541. $9{,}261 \times 3$

542. $48 \cdot 76$

543. $64 \cdot 10$

544. $1{,}000(22)$

545. 162×493

546. $(601)(943)$

547. $3{,}624 \times 517$

548. $10{,}538 \cdot 22$

Translate Word Phrases to Math Notation

In the following exercises, translate and simplify.

549. the product of 15 and 28

550. ninety-four times thirty-three

551. twice 575

552. ten times two hundred sixty-four

Multiply Whole Numbers in Applications

In the following exercises, solve.

553. **Gardening** Geniece bought 8 packs of marigolds to plant in her yard. Each pack has 6 flowers. How many marigolds did Geniece buy?

554. **Cooking** Ratika is making rice for a dinner party. The number of cups of water is twice the number of cups of rice. If Ratika plans to use 4 cups of rice, how many cups of water does she need?

555. **Multiplex** There are twelve theaters at the multiplex and each theater has 150 seats. What is the total number of seats at the multiplex?

556. **Roofing** Lewis needs to put new shingles on his roof. The roof is a rectangle, 30 feet by 24 feet. What is the area of the roof?

Divide Whole Numbers

Use Division Notation

Translate from math notation to words.

557. $54 \div 9$

558. $42/7$

559. $\frac{72}{8}$

560. $6\overline{)48}$

Model Division of Whole Numbers

In the following exercises, model.

561. $8 \div 2$

562. $3\overline{)12}$

Divide Whole Numbers

In the following exercises, divide. Then check by multiplying.

563. $14 \div 2$

564. $\frac{32}{8}$

565. $52 \div 4$

566. $26\overline{)26}$

567. $\frac{97}{1}$

568. $0 \div 52$

569. $100 \div 0$

570. $\frac{355}{5}$

571. $3828 \div 6$

572. $31\overline{)1{,}519}$

573. $\frac{7505}{25}$

574. $5{,}166 \div 42$

Translate Word Phrases to Math Notation

In the following exercises, translate and simplify.

575. the quotient of 64 and 16

576. the quotient of 572 and 52

Divide Whole Numbers in Applications

In the following exercises, solve.

577. **Ribbon** One spool of ribbon is 27 feet. Lizbeth uses 3 feet of ribbon for each gift basket that she wraps. How many gift baskets can Lizbeth wrap from one spool of ribbon?

578. **Juice** One carton of fruit juice is 128 ounces. How many 4 ounce cups can Shayla fill from one carton of juice?

Practice Test

579. Determine which of the following numbers are
ⓐ counting numbers
ⓑ whole numbers.
$0, 4, 87$

580. Find the place value of the given digits in the number 549,362.
ⓐ 9 ⓑ 6 ⓒ 2 ⓓ 5

581. Write each number as a whole number using digits.
ⓐ six hundred thirteen
ⓑ fifty-five thousand two hundred eight

582. Round 25,849 to the nearest hundred.

Simplify.

583. $45 + 23$

584. $65 - 42$

585. $85 \div 5$

586. $1{,}000 \times 8$

587. $90 - 58$

588. $73 + 89$

589. $(0)(12{,}675)$

590. $634 + 255$

591. $\frac{0}{9}$

592. $8\overline{)128}$

593. $145 - 79$

594. $299 + 836$

595. $7 \cdot 475$

596. $8{,}528 + 704$

597. $35(14)$

598. $\frac{26}{0}$

599. $733 - 291$

600. $4{,}916 - 1{,}538$

601. $495 \div 45$

602. 52×983

Translate each phrase to math notation and then simplify.

603. The sum of 16 and 58

604. The product of 9 and 15

605. The difference of 32 and 18

606. The quotient of 63 and 21

607. Twice 524

608. 29 more than 32

609. 50 less than 300

In the following exercises, solve.

610. LaVelle buys a jumbo bag of 84 candies to make favor bags for her son's party. If she wants to make 12 bags, how many candies should she put in each bag?

611. Last month, Stan's take-home pay was $\$3{,}816$ and his expenses were $\$3{,}472$. How much of his take-home pay did Stan have left after he paid his expenses?

612. Each class at Greenville School has 22 children enrolled. The school has 24 classes. How many children are enrolled at Greenville School?

613. Clayton walked 12 blocks to his mother's house, 6 blocks to the gym, and 9 blocks to the grocery store before walking the last 3 blocks home. What was the total number of blocks that Clayton walked?